SEQUOIA

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Simulated **E**valuation of **QU**antum **O**ptics **I**nstruments and **A**nalysis
Build a lab setup from characterized parts, then predict beams, counts, CAR, g2, HOM and CHSH.

1 · FROM PARTS TO PREDICTIONS

Parts library — part.yaml + CSV, provenance per value
→ **Setup** — graph on a grid, free-space · fiber
→ **Validation** — port kinds · cycles, units · dangling ports
→ **Classical engine** — Jones · Gaussian q, power · pol. · size
→ **Rate engine** — biphoton amplitude, detectors · tagger
→ **Readouts** — singles · CAR, g2 · HOM · CHSH

PySide6 GUI or plain YAML

one linear network, two statistics layers

GUI · CLI

2 · COMPONENT MODEL

Category — **Detectors**
palette group, port conventions

Type — **SNSPD**
Python class + pydantic schema (physics)

Part — **snspd_generic**
data only: scalars, curves, surfaces

Instance — **spad_h @ (34, 6), rot 90°**
settings + overrides in a setup file

Value resolution: instance override › measured › datasheet › model › ideal default. New part = data only; new type = one Python file.

Links carry the physics: free-space (length, medium), fiber (part, length, connector), cable (delay). Ideal ↔ characterized toggle gives a loss budget.

3 · TWO LAYERS ON ONE NETWORK

CLASSICAL
Jones 2×2 per wavelength: loss, polarization, phase
Gaussian q-parameter (ABCD, M²): w, R, waist, fiber overlap
coherent paths from one source: coherence × mode overlap

pump state at the crystal

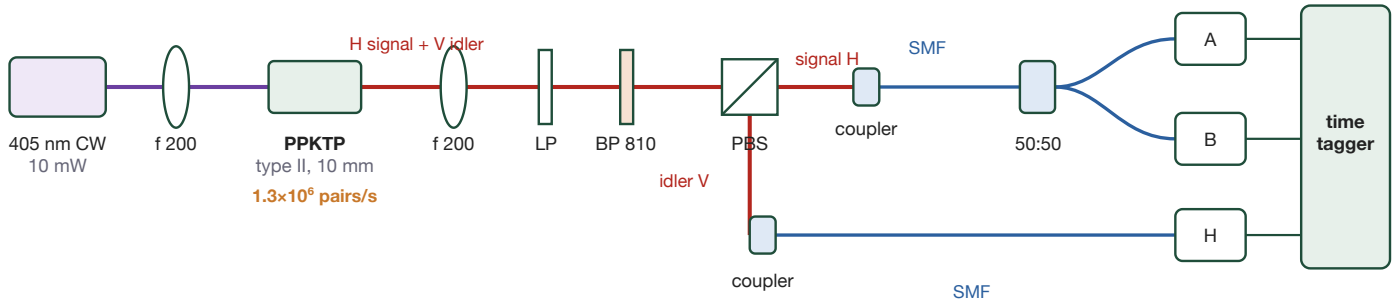
QUANTUM (low gain, first order in pair rate)
pair source: biphoton amplitude $F(\omega_1,\omega_2)$ × pair rate (spdcalc)
same Jones + delays → detector POVM, dead time, jitter, window
singles, coincidences, accidentals, g2, HOM, tomography

$$A_{AB} = J_{A\leftarrow s} \otimes J_{B\leftarrow i}\, F(\omega_1,\omega_2) + J_{A\leftarrow i} \otimes J_{B\leftarrow s}\, F(\omega_2,\omega_1)$$

$$P_{AB} = \iint A\dagger\, (D_A \otimes D_B)\, A$$ (D: detector efficiency)

One formula covers heralding, HBT, HOM (exchange term), Bell tests, tomography.

4 · WORKED EXAMPLE: HERALDED HBT WITH TYPE-II PPKTP (`setups/hbt_heralded.yaml`)



Pairs emitted into collection modes	**1.3 × 10⁶ s⁻¹**
Herald singles (H)	**6.0 × 10⁵ s⁻¹**
True coincidences H×A, 1 ns window	**1.3 × 10⁵ s⁻¹**
CAR (C_true / accidentals)	**736**
Heralded g2(0)	**0.0027**

Generic placeholder parts, ideal alignment. Remove the long-pass filter and the pump-leakage diagnostic fires, with the count rate it adds to each detector. Replace parts by measured ones (curves, DCR, jitter) to move from ideal to characterized.

VALIDATED AGAINST

- Malus, HWP rotation, thin lens, telescope, Gaussian overlap
- MZI visibility = $|\gamma(\tau)|$; CHSH $S = 2\sqrt{2}\cdot V$; unheralded $g2 = 1+1/K$
- HOM curve vs spdcalc; dead time vs Monte Carlo
- 777 tests; three independent review rounds on the quantum layer

KNOWN LIMITS

- Absolute SPDC rate uncertain (×0.4 to ×2): use a measured *brightness*
- Collection modes are part data, not back-propagated
- Thin lenses, mode at centre wavelength, no GDD/PMD
- Feed-forward only: no cavities, no back-reflections

ROADMAP

M0–M5 done: model, engines, GUI, analysis
M6 library manager
M7 Monte-Carlo time tags
M8 Gaussian multimode

Stack: Python 3.12, numpy/scipy, pint, pydantic, spdcalc, PySide6, pyqtgraph.